

CO1 & CO2 Practice questions

1. A transportation department gathers data on vehicle speed, road width, and accident frequency at major intersections. The department aims to study relationships among traffic characteristics. Use covariance analysis to explore how traffic variables are related and interpret the findings to improve traffic management and road safety planning. Also write the code to apply performance tuning using .loc on the significant indicators and then calculate the covariance matrix using python.

Int	Speed (X_1)	Width of Road (X_2)	Accidents (X_3)
1	60	12	5
2	55	10	7
3	50	9	9
4	65	14	4
5	45	8	11
6	70	15	3
7	52	9	8
8	48	8	10

2. The government tests three pet protection programs to measure improvement scores of rescued pets. Each program evaluates 10 pets. Program A (Vaccination Program), Program B (Shelter & Nutrition Program) and Program C (Medical Care Program) is done. Using one-way ANOVA at 5% significance level, test whether there is a significant difference in the effectiveness of the government pet protection programs.

Program A	Program B	Program C
68	78	58
70	80	60
72	82	62
69	79	59
71	81	61
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3. A software testing team claims that their new compiler optimization reduces the average program execution time to 50 milliseconds. To test this claim, 8 benchmark programs are executed using the optimized compiler. The execution times (in ms) are: 48, 52, 47, 51, 49, 53, 46, 50. At 1% significance level, test whether the average execution time is significantly different from 50 ms using T-test.
4. A researcher wants to find out whether the average regular gasoline price is higher than \$2.45 in the Mid-West region. The sample analysed of 25 observations shows \$2.65 with standard deviation of \$0.35. State the hypothesis and at 95% confidence level, determine if the claim can be accepted or not using Z-test.
5. A clinical researcher compares Tablet A and Tablet B for reducing fever (measured in °F after 24 hours). Tablet A is tested with 10 patients reducing the fever by 3.1, 2.8, 3.4, 3.0, 2.9, 3.3, 3.2, 2.7, 3.0, 3.1. Tablet B is tested with 10 patients reducing the fever by 2.5, 2.7, 2.4, 2.6, 2.8, 2.3, 2.6, 2.5, 2.4, 2.7. At $\alpha = 0.05$, test whether there is a significant difference between the tablets using T-test.
6. The random variables X and Y have $\text{Var}(X) = 9$ and $\text{Var}(Y) = 4$ and $\text{Corr}(X, Y) = -1/6$. Determine $\text{Var}(X-3Y+4)$.
7. Differentiate between Covariance and Correlation. Determine the covariance(X,Y) from the given data as correlation(X,Y) is 0.6 and $\sigma_x=5$ and $\sigma_y=4$.
8. A library system lends books for periods of 21 days. This policy is being reevaluated in view of a possible new loan period that could be either longer or shorter than 21 days. To aid in making this decision, book-lending records were consulted to determine the loan periods actually used by the patrons. A random sample of eight records revealed the following loan periods in days: 21, 15, 12, 24, 20, 21, 13, and 16. Test the null hypothesis with t, using the .05 level of significance.
9. A psychologist investigates the effect of instructions on the time required to solve a puzzle. Each of 20 volunteers is given the same puzzle to be solved as rapidly as possible. Subjects are randomly assigned, in equal numbers, to receive two different sets of instructions prior to the task. One group is told that the task is difficult (X1), and the other group is told that the task is easy (X2). The score for each subject reflects the time in minutes required to solve the puzzle. Use a t to test the null hypothesis at the .05 level of significance .

SOLUTION TIMES	
"DIFFICULT" TASK	"EASY" TASK
5	13
20	6
7	6
23	5
30	3
24	6
9	10
8	20
20	9
12	12

10. A number of investigators have reported a tendency for more people to die (from natural causes, such as cancer and strokes) after, rather than before, a major holiday. This post-holiday death peak has been attributed to a number of factors, including the willful postponement of death until after the holiday, as well as holiday stress and post-holiday depression. Writing in the Journal of the American Medical Association (April 11, 1990), Phillips and Smith report that among a total of 103 elderly California women of Chinese descent who died of natural causes within one week of the Harvest Moon Festival, only 33 died the week before, while 70 died the week after.

- (a) Using the .05 level of significance, test the null hypothesis that, in the underlying population, people are equally likely to die either the week before or the week after this holiday.
- (b) Specify the approximate p-value for this test result.
- (c) How might this result be reported in the literature?